SOP-DESIGN PROCESS

[CONFIDENTIAL]
[UTTHUNGA]
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CHANGE HISTORY

SL. No.	Document Version No.	Change Description	Reason For change	Prepared By & Date	Reviewed By & Date	Approved By & Date
01	0.1	Draft	Implementation of the Integrated Management System	Sushmitha S 04-Jul-22		
02	Issue B Version 1.0	Revised the Document to suit the requirements for implementation of the integrated management system			Megha VK 06-Jul-22	Rama G 08-Jul-22
03	Issue B Version 1.1	Revisited the Design	Added the secure development lifecycle	Sachin A 20-Jul-23	Santosh V 24-Jul-23	Rama G 26-Jul-23
04	Issue B Version 1.2	Document formatting, font style to Poppins and alignment	Updated in line with the document guidelines	Vanavee P 20-Dec-24	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
5	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-Aug-25	Santosh V 08-Aug-25	Sanmangala M S 14-Aug-25

This section describes the details of changes that have resulted in the current document.

1 Purpose

To establish a standard method for preparing, reviewing and approving the Design documents.

2 Scope

This applies to all the projects executed by Utthunga.

3 Abbreviations

a. PM: Project Manager

b. RS: Requirements Specification

c. TST: Testing

d. DAR: Decision Analysis and Resolution

4 Terms and Definitions

NIL.

5 Responsibilities

a. Execute full software development life cycle (SDLC)

b. Gather the project specific requirements

c. Handle the Requirement change

d. Use configuration management

e. Review the requirements

f. Take approval from customer

g. Verify and validate the requirements

h. Selecting appropriate design approach and standards

6 RACI

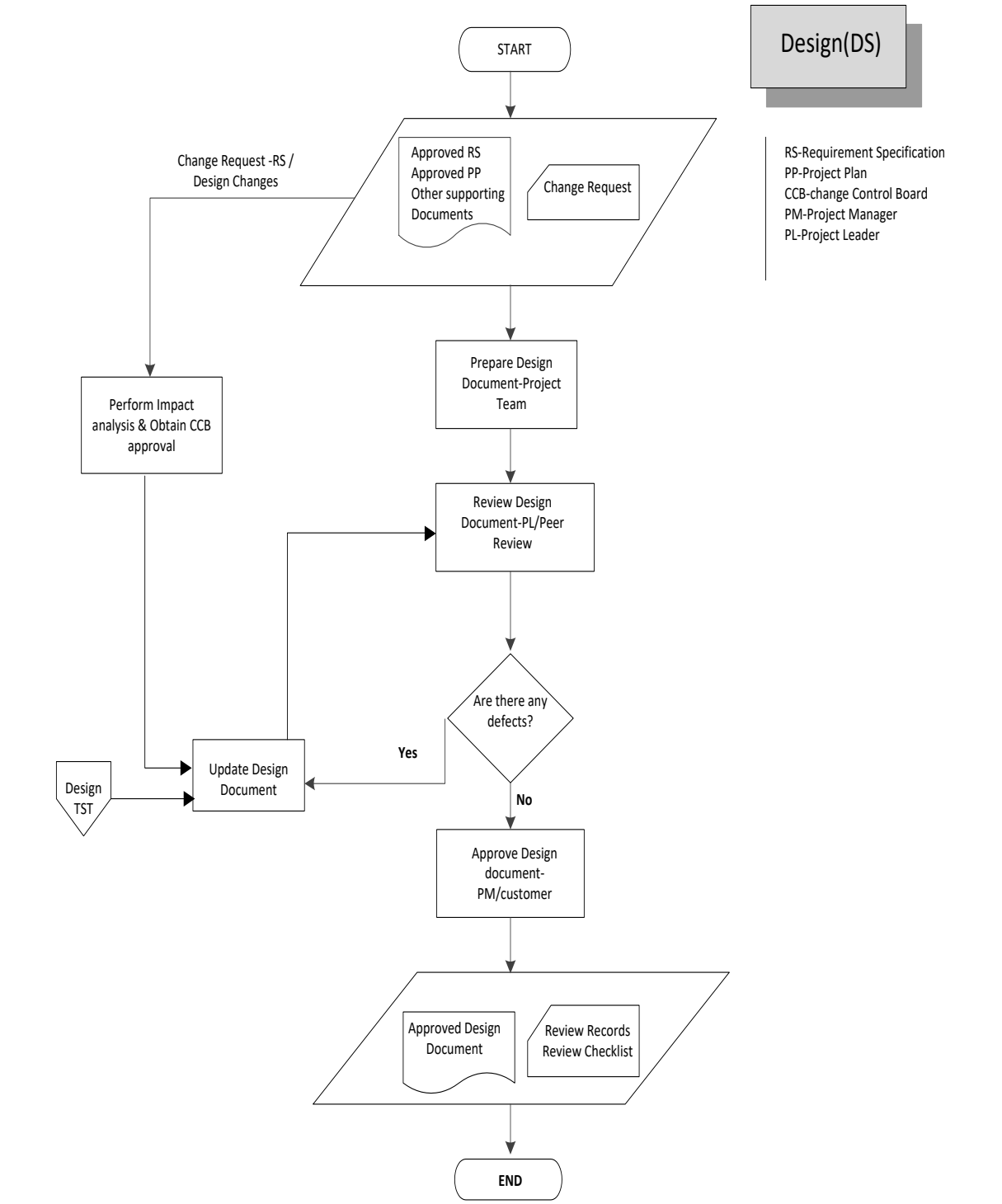
Process	Responsible	Accountable	Consulted	Informed
Design process	Technical Lead/ Architect	Project Manager	Architect	Customer / Delivery Manager

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Customer / Internal Process / Quality	Approved Requirements Specification (RS), approved Project Plan, Change Request and any other supporting documents	Design process	Assessment, Review and approval Process	Approved Design Document	Customer / Internal Process

8 Overall Process Flow Chart

a. Design Process



9 Process Description

9.1 Design Process

9.1.1 Design Planning

a. The Project Manager or the person designated by Project Manager shall perform Design and prepare the Design document in line with the Requirements Specification (RS).

b. The Design shall be prepared as per the Design Template, selecting appropriate design approach and standards.

c. The Design shall make appropriate references to the RS to ensure all the requirements are addressed and are traceable.

d. Incase of Security requirements in scope , Secure Design Principles to be followed

e. Identify Security Vulnerabilities, determine Risk from those threats as per the Threat Modeling guidelines available in IMS

f. The Design shall be reviewed and approved by the Project Manager or person designated by Project Manager and the Customer.

g. The Requirement Traceability shall be updated with design references.

h. Information architecture, system architecture, Blueprint, interface design and various high level designs shall be developed based on the chosen architectural decision, past experience, integration strategy with external & internal applications etc.

9.1.2 Design Description

a. This shall be developed based on the chosen architectural decision, experience, integration strategy with external & internal applications etc.

b. The technical architect shall evaluate alternative architecture options along with the team.

c. A formal evaluation process shall be followed developing architecture and design for new products and major enhancements.

d. For Selecting and evaluating the design, refer DAR process

9.1.3 Design Input

a. The Project Manager or person designated by Project Manager shall ensure that the inputs to the systems are as per the input requirements mentioned in the RS.

b. Any ambiguous or conflicting requirements shall be resolved by the Project Manager or person designated by Project Manager. Wherever need the Project Manager or person designated by Project Manager shall contact the Customer for clarifications and amendments.

9.1.4 Design Controls - Review, Verification, Validation and Approval

a. The Project Manager (PM) or person designated by PM shall maintain the documents and quality records related to design in the project file (hardcopy) or as soft copy in the configuration area. Following documented information shall be maintained

b. Design Document

c. Review Checklist

d. Review Records

e. Review shall be done as described in the Review Process.

f. Design Review Checklist shall be used in the review of the design document.

g. Customer may approve the Design as required prior to proceeding with the subsequent phase of the project.

9.1.5 Design Output

a. The Project Manager or person designated by Project Manager shall ensure that the design covers all the requirements mentioned in the RS.

b. The Project Manager or person designated by Project Manager shall ensure that the design output

c. Meets the design output requirements

d. Identify design characteristics that are crucial to the safe and proper functioning.

9.1.6 Design Change

a. Any changes identified by the Customer or Utthunga Technologies shall be reviewed and approved by the Project Manager or Customer prior to implementation as defined in the Configuration Management process.

10 Information Security requirements

An access control policy shall be established, documented, and reviewed based on business and information security requirements. Users shall only be provided with access to the network and network services that they have been specifically authorized to use

a. Access to program source code shall be restricted
b. Password management systems shall be interactive and shall ensure quality passwords.
c. Changes to the organization, business processes, information processing facilities and systems that affect information security shall be controlled.
d. Development, testing, and operational environments shall be separated to reduce the risks of unauthorized access or changes to the operational environment.
e. Backup copies of information, software and system images shall be taken and tested regularly in accordance with an agreed backup policy.
f. Event logs recording user activities, exceptions, faults and information security events shall be produced, kept and regularly reviewed.
g. System administrator and system operator activities shall be logged and the logs protected and regularly reviewed.
h. The clocks of all relevant information processing systems within an organization or security domain shall be synchronized to a single reference time source
i. Information about technical vulnerabilities of information systems being used shall be obtained in a timely fashion, the organization's exposure to such vulnerabilities evaluated and appropriate measures taken to address the associated risk.
j. Rules governing the installation of software by users shall be established and implemented.
k. Audit requirements and activities involving verification of operational systems shall be carefully planned and agreed to minimize disruptions to business processes.
l. Formal transfer policies, procedures and controls shall be in place to protect the transfer of information using all types of communication facilities.
m. Agreements shall address the secure transfer of business information between the organization and external parties.
n. Information involved in electronic messaging shall be appropriately protected.
o. The information security related requirements shall be included in the requirements for new information systems or enhancements to existing information systems.
p. Information involved in application services passing over public networks shall be protected from fraudulent activity, contract dispute and unauthorized disclosure and modification.
q. Information involved in application service transactions shall be protected to prevent incomplete transmission, misrouting, unauthorized message alteration, unauthorized disclosure, unauthorized message duplication or replay.
r. Confidentiality of data is maintained

11 Metrics

NIL

12 References

a. ISO 9001 Clauses 8.3

b. ISO 27001 Clauses 8.1, Controls A.6, A.7, A.9, A.10, A.12, A.14

c. ISMS Control Policies

d. Secure Design Principles UT/SD/DSP/GDL/01

e. Security Guidelines UT/IEC/GDL/04

f. Threat Modeling Guidelines UT/SD/DSP/GDL/02

g. Vulnerability Handling Guidelines UT/SD/DSP/GDL/03

13 Annexures

NIL

14 Forms

a. Design Document Template UT/SD/DSP/TMP/01

b. Design Document Review Checklist UT/SD/DSP/TMP/02

c. Security Architecture and Design Review Checklist UT/SD/DSP/TMP/03